Contents

The 20-minute talk **1**
What the deck argues . . . 1
Where the content comes from . . . 2
The documents as PDF . . . 2
House style . . . 3
Rebuilding . . . 3
When S15 lands . . . 3

The S16 deck: `Hunter_SST_second_reference.pptx` **4**
Files . . . 4
Rebuild . . . 4
Claims this deck deliberately qualifies . . . 5

The 20-minute talk

`Hunter_SST_techniques.pptx` is the current deck — 34 slides (31 presented, 3 backup), ~22 min as written (the script's cut-list lands it at 20), click-built, and current as of the S14sam3 / S14sec bundles. `Hunter_SST_talk_handbook.pdf` combines the two documents below into one printable file. `TECHNICAL_QA.md` answers the "how did you implement this" questions at the level of functions, formulas, thresholds and guards, ordered so sections 1–3 back slide 32, 4–8 back slide 33, and 9–12 back slide 34. Read it before presenting; it is what you reach for when a question goes past the slide.

`NARRATION.md` is the speaking guide, not a script: per slide it gives the time to spend, what the slide has to establish, which numbers to say out loud, and what to leave for questions. The deck's presenter notes carry the same directions in one short paragraph each. `Hunter_SST_techniques.pdf` is the deck flattened (all build steps visible at once) for handout or projector fallback.

`Hunter_SST_OC-CCL_findings.*` is the earlier cut, kept for reference. It told the reproduction story and spent two slides on paper-vs-release discrepancies; the current deck drops that thread entirely and walks the techniques instead.

What the deck argues

One baseline — uniform supports, no technique, Beetle 55.96 ± 7.90 and Butterfly Major 68.54 ± 3.15 — then each technique added to it and measured the same way: 20 seeded draws, the same draws for both arms, paired per seed, bootstrap CI, band frozen before the run. Slide 24 is the ledger those measurements add up to, and slide 3 defines the unit they are all in: mIoU per trait present in an image's ground truth, averaged over traits, then images, then the 20 runs.

Structure:

slides	part
1–2	the task
3	**the metric** — how one image is scored, and what a number means
4	the baseline everything else is measured against
5–9	support-side techniques: curation, round-robin balancing, placement, more shots
10–11	the query-side defect that support-side techniques cannot reach
12–16	**orientation TTA** — mechanism (2 slides), results, and the redundancy warning

slides	part
17–19	the loss: the cycle and its two preconditions
20	SCNP — the borrowed topology term, tried twice, closed twice
21–22	every loss variant, and the oracle ceiling that explains them
23–25	scoring as a technique, the ledger, where the numbers land
26–29	**the model swaps** — SeC-4B (doesn't pay), SAM 3 (does), and why: the ceiling moved
30–31	the pre-registered 5-shot projection, close
32–34	**backup, not presented** — how canonicalization is implemented, and what guards it

The last act is the S14 result. SeC-4B loses under the deployment stack (−2.79 ✱ / −3.02 ✱) and its concept injection costs ~6 points; SAM 3 pays on beetle in four composing rungs (backbone +2.14 ✱, TTA +0.99 ✱, our loss stack +1.59 ✱, the new architecture arm +1.96 ✱), taking beetle's best 1-shot number from 64.12 to **69.88** (+5.77 ✱). The mechanism slide is the one that matters: SAM 3's assignment pot is *smaller* than SAM 2's (1.80 vs 2.65), so the gains are not assignment-bounded — the oracle ceiling itself moved, 66.76 → 71.08, because the arch arm trains the proposal-forming module. Butterfly is flat across both families, confirming its residual is protocol, not capacity.

TTA gets the deepest treatment: `rotate_episode` (image and labelmap rotated together, query untouched, its own `@r180` cache key), the decision rule (`top[top > 0].mean()`, −inf when nothing is claimed, no ground truth consulted), what it costs (2× inference, kept at 2× rather than ~10× by caching both orientations), what it bought (+5.44 / +4.88 butterfly, +2.04 beetle at 5-shot, cyrbia 47.40 → 73.27), and what it costs in variance (cyrbia per-run spread 6.81 → 10.89).

Where the content comes from

Nothing is drawn or invented. Every figure is either

- **a render this project already produced** — `results/s3/overlay.png`, cropped into its five panels by `make_talk_figures.py::f11_overlay_panels`, and rotated 180° for the TTA mechanism slide by `f14_rotated_panels`, which is the same `np.rot90(·, 2)` the adapter applies; or
- **a plot of numbers lifted live from a landed bundle** in `results/` and `results/inbox/` — and where a README constant and a bundle value both exist, the script `asserts` they agree before plotting (`close()`).

Shapes are used only to frame, number and point. There is no generated imagery.

The documents as PDF

`./docs_to_pdf.sh` renders everything to PDF beside the sources, via pandoc and xelatex. xelatex rather than pdflatex because the documents contain the significance mark, arrows, degree signs and a true minus sign. Re-run it after editing any markdown here.

`Hunter_SST_talk_handbook.pdf` is the one to print: the speaking guide and the technical Q&A in a single document, with a contents page that lists every slide entry and every question. `make_handbook.py` builds it by demoting each source document's headings one level under a part heading, so the two stay authored separately and combine on render. The individual `NARRATION.pdf` and `TECHNICAL_QA.pdf` are still produced if you want one without the other.

House style

The deck copies the visual system of `Hunter_Bridges_Introduction.pptx`:

- **Courier New throughout**, at every level. Titles 30pt bold, eyebrow 12pt bold uppercase with 2pt tracking, body 11 to 14pt.
- **Black on white, with one deliberate exception.** Photographs go through luminance (`make_talk_figures.py::_to_grey`). **Label maps keep their colour**, because the hue *is* the trait identity: head, pronotum, elytra, antennae and legs are only distinguishable by it, and a grey ramp cannot carry five categories at slide scale. The rule the audience can infer: colour on a slide always means "this encodes a trait". Charts, chrome and type stay monochrome.
- **Square boxes**, white fill, 1.75pt black outline. No rounded corners, no tints, no shadows.
- **A full-width black rule under every title**, at y=1.62in, exactly as the reference deck sets it. Nothing may sit on or above it; `build_deck.py` clamps figures, pictures and cards to y ≥ 1.95in so this cannot regress.
- Charts follow: black fill for the subject, mid grey for a comparator, white with a black edge for a negative, hatching for a pre-registered band. Sign reads from which side of the zero line a bar sits on, not from colour.

Courier is about 25% wider than a proportional face, so the type scale was rebuilt for it and titles are kept to one line. `check_overlaps.py` measures in DejaVu Sans Mono for this reason.

Slide text is plain and technical: statements of fact with the numbers carrying the weight, no aphorisms, and **no em-dashes anywhere** (slides, figures, presenter notes or the speaking guide). `retone.py` records the rewrite that established this, and `geo_fix.py` the geometry pass that followed the restyle. Check with:

```
python -c "from pptx import Presentation; d=Presentation('Hunter_SST_techniques.pptx'); \
print(sum('—' in sh.text_frame.text for s in d.slides for sh in s.shapes if sh.has_text_f
```

Rebuilding

```
cd docs/talk
python make_talk_figures.py --out figures            # figures from results/
python build_deck.py                                 # writes the .pptx + anim_spec.json
python add_animations.py Hunter_SST_techniques.pptx anim_spec.json \
       Hunter_SST_techniques_anim.pptx               # injects the <p:timing> builds

python check_overlaps.py Hunter_SST_techniques_anim.pptx deck.pdf   # layout sweep
```

`check_overlaps.py` is the layout regression check — run it after any edit. It measures each text block's real rendered height with font metrics, then flags text that spills onto the shape below it, text crossing a card boundary, and (from the exported PDF's word boxes) any two lines whose ink actually overlaps. One entry is a known false positive and stays in the output: slide 18's "prompt back" label sitting above its arrow, verified by eye. The edge rows are the 0.33in eyebrow labels, which is deliberate.

`python-pptx` has no animation API, so `add_animations.py` writes the OOXML timing tree directly: one fade entrance per shape, grouped into clicks by `anim_spec.json` (`{slide_index: [[shape_id, ...], ...]}`). Slides 7, 21 and 25 build a chart by stacking two pixel-identical PNGs and revealing the later one — which is why those figures are saved with `tight=False`.

When S15 lands

The 5-shot run on the champion is the next thing that changes this deck.

0. Backup slides 32–34 are static text and do not need rebuilding unless `src/occl/curation.py` changes; if it does, re-read it against `TECHNICAL_QA.md`.
1. **Slide 30** (`f22_projection.png`) — the projection bands become measurements. Edit `make_talk_figures.py::f22_projection` to plot the landed values beside the bands rather than replacing them; a projection that was written down and then scored is worth more than a bar.

2. **Slide 28** — if 5-shot beats it, the "beetle's best 1-shot number" line becomes a best-overall line.
3. **Slide 24** — a new ledger row ("more shots on SAM 3"), which will re-sort the figure.
4. **Slide 31** — the closing lines stand unless the projection misses its band.

`f10_s14.png` (the pre-registration bands) is no longer on a slide — the results superseded it. It stays in `figures/` in case the write-up wants the pre-registration shown beside the outcome.

If a bundle disagrees with the pins the deck quotes (76.52 / 64.12 for the SAM 2 crown, 69.88 / 76.64 for the SAM 3 arms), the gate fails before the deck is wrong: `check_gate.py s14sam3 --bundle s14sam3_<ds>.zip`.

The S16 deck: `Hunter_SST_second_reference.pptx`

A separate, shorter talk on one stage: which second reference pays, the implicit one (orientation TTA's rotated support) or the explicit one (a real 2-shot). It does not replace `Hunter_SST_techniques.pptx` and shares none of its files beyond the animation injector and the layout checker.

19 slides, 17 presented and 2 backup, about 13 to 14 minutes.

Files

file	what it is
`make_s16_figures.py`	every figure, from `results/s16.zip` and the named README rows, into `figures_s16/`
`build_s16_deck.py`	the deck, writing `anim_spec_s16.json` alongside it
`s16_factcheck_fixes.py`	the recorded corrections from the fact-check pass
`verify_s16.py`	checks every decimal and percentage in the deck PDF and both markdown files against the record
`NARRATION_S16.md`	speaking guide, directions per slide
`TECHNICAL_QA_S16.md`	implementation answers, what backup slides 18 and 19 point at
`make_handbook_s16.py`, `docs_to_pdf_s16.sh`	the combined handbook and the PDF renders

Rebuild

```
python make_s16_figures.py
python build_s16_deck.py
python add_animations.py Hunter_SST_second_reference.pptx \
    anim_spec_s16.json Hunter_SST_second_reference.pptx
soffice --headless --convert-to pdf --outdir . Hunter_SST_second_reference.pptx
python check_overlaps.py Hunter_SST_second_reference.pptx \
    Hunter_SST_second_reference.pdf
./docs_to_pdf_s16.sh
python verify_s16.py
```

Same house style as the main deck: Courier New, black on white, the rule at 1.62in, colour only where a label map's hue carries trait identity, and no generated imagery. The only photographs are panels of `results/s3/overlay.png`, with the implicit arm's second panel being that support rotated 180 degrees.

Claims this deck deliberately qualifies

The fact-check pass caught six overreaches inherited from the README's S16 result section, and the deck now states each the careful way. If the README is edited later, keep these:

1. The two 1-shot TTA cells **resume** S11's banked cells (identical evaluator fingerprint, zero inference, bit-identical values). They certify the wiring of the merge, not TTA. The 2-shot cells are the only new measurement.
2. `best-of-3` is `miou_boundary.independent_best3`, so it changes the overlap rule **and** the boundary tolerance, not the boundary alone.
3. Beetle's share of the 1 to 5 shot lift, 58.6%, is **past** the top of its pre-registered 35 to 55% band. Butterfly undershot, beetle overshot.
4. 0.30 is the tightest cell **at 1 or 2 shots**; S6's Major 5-shot cells sit at 0.15.
5. Two labels is the cheapest **training-free** point above the paper's 65.2 row. S14sam3's arch_tta is above it at one label.
6. A second label is the strongest **per-label** lever on beetle; 5-shot is larger in absolute terms (+9.96).

Also: the oracle reading is not paired, so ceiling moves are differences rather than tested effects, and +5.84 and +1.18 are unpaired against banked 1-shot levels because those cells were deliberately not re-run.